A GROUPOID SCHEME OVER THE SPECTRUM OF A FIELD HAS SEPARATED ARROW SCHEME

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Allowed standard prerequisites (editorial).

- The stabilizer group scheme of a groupoid over the spectrum of a field is separated (Groupoids chapter, Lemma groupoids-lemma-group-scheme-over-field-separated, canonical tag 047L). cited at proof lines 1177-1179. Gives separatedness of G to U, the input to the diagonal lemma.
- The diagonal j = (t, s) of a groupoid is separated as soon as the stabilizer is separated (Groupoids chapter, Lemma groupoids-lemma-diagonal, canonical tag 02YI). cited at proof lines 1179-1181. Yields that j = (t, s) : R to U x_S U is separated.
- Fibre products over the spectrum of a field are affine, by the affine-local construction of fibre products (Schemes chapter, Section schemes-section-fibre-products, canonical tag 01JO; kind reference). cited at proof lines 1182-1184. Gives that U x_S U is affine when U = Spec(k), making the separated target affine.
- Separated over affine implies separated: permanence of separatedness (Schemes chapter, Lemma schemes-lemma-separated-permanence, canonical tag 01KU). cited at proof lines 1185-1186. Concludes from j separated with affine target that R is a separated scheme.

Source reference. Original lemma 047L: sources/stacks-chow/tags.txt, source lines 5512–5512. Cited at proof lines 1177-1179: the stabilizer group scheme G to U is separated.

Source reference. Original lemma 02YI: sources/stacks-chow/tags.txt, source lines 3877–3877. Cited at proof lines 1179-1181: the morphism j = (t, s) : R to U x_S U is separated.

Source reference. Original reference 01JO: sources/stacks-chow/tags.txt, source lines 2021–2021. Cited at proof lines 1182-1184: U x_S U is affine by the construction of fibre products when U is the spectrum of a field.

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Source reference. Original lemma 01KU: sources/stacks-chow/tags.txt, source lines 2063–2063. Cited at proof lines 1185-1186: R separated over the affine scheme U x_S U is a separated scheme.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. If U is the spectrum of a field, then R is a separated scheme.*

Proof. By Groupoids, Lemma 047L the stabilizer group scheme $G \to U$ is separated. By Groupoids, Lemma 02YI the morphism $j = (t, s) : R \to U \times_S U$ is separated. As U is the spectrum of a field the scheme $U \times_S U$ is affine (by the construction of fibre products in Schemes, Section 01JO). Hence R is a separated scheme, see Schemes, Lemma 01KU. □